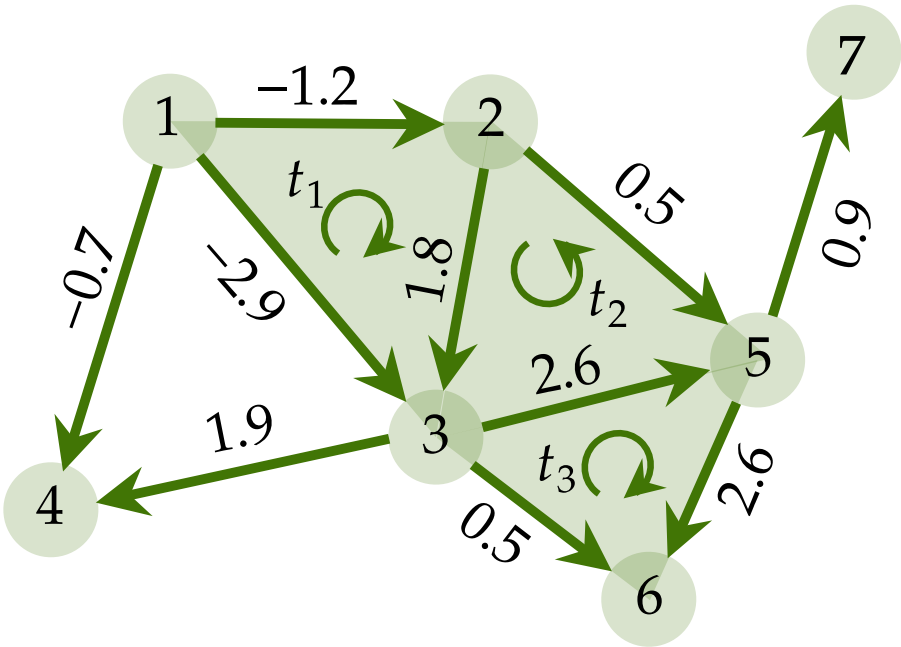
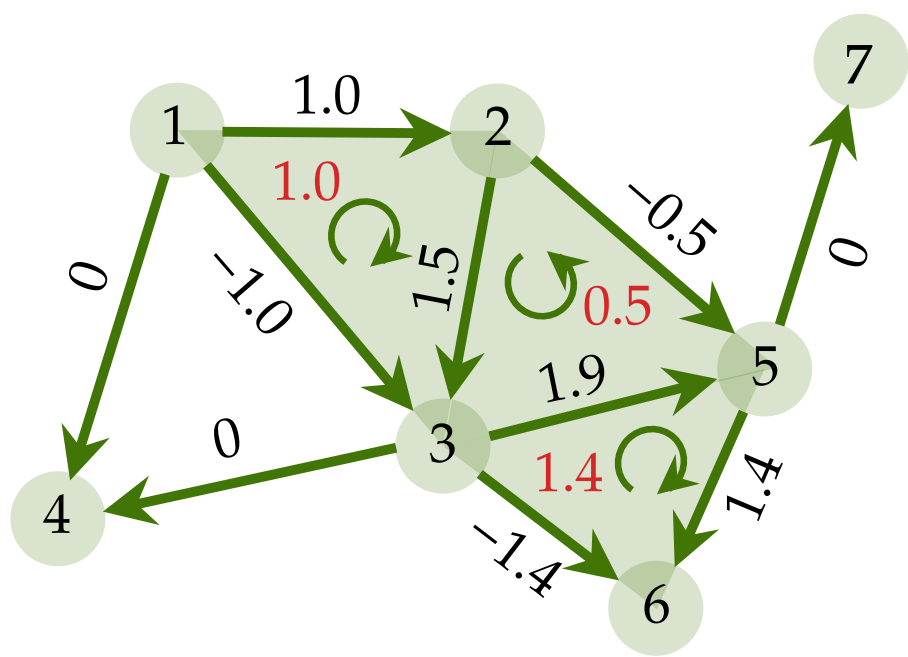
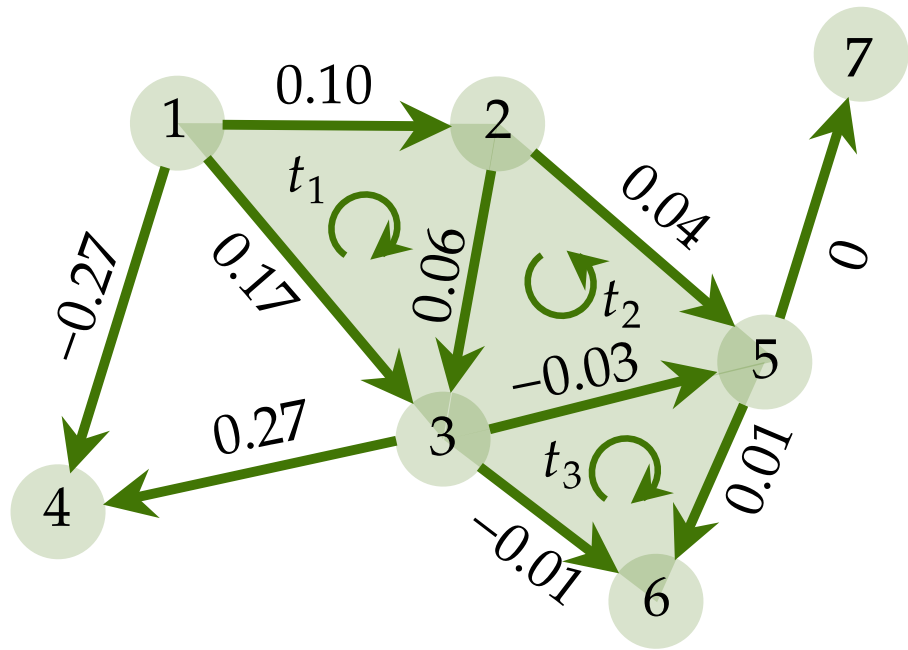


Hodgdenproposition

Lovászal.2004;Linetal.2020

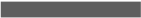












Gradient flow

Curl-free, irrotational

Harmonic flow

Div- and curl-free

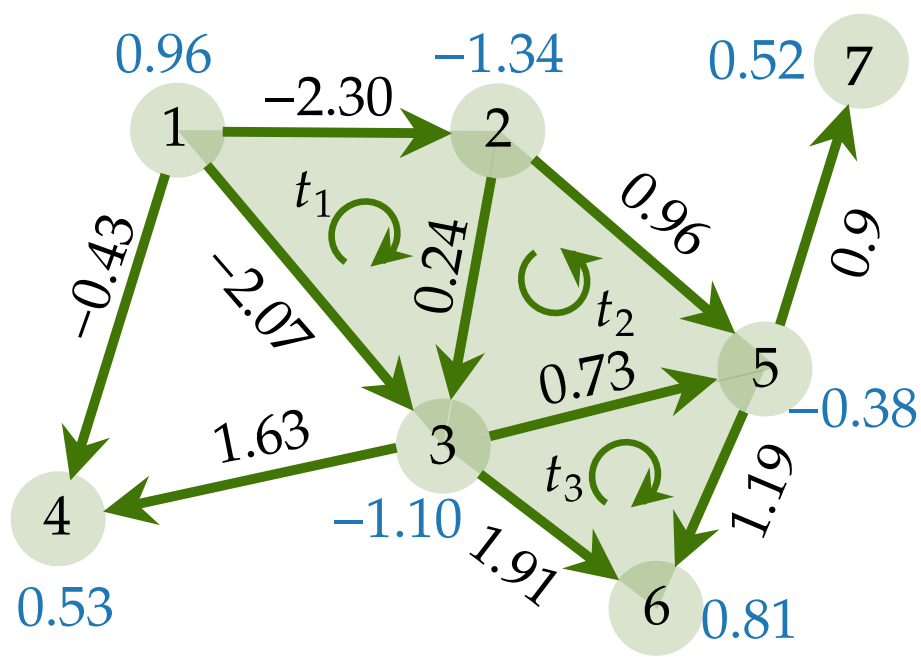
Curl flow

Div-free, solenoidal



- This holds for the space of nodes, edges, triangles ... and **almost everywhere**

- What is the case for nodes? The harmonic space $\sim$ #connected components





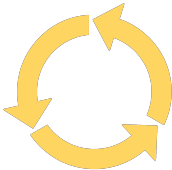
$$\mathbb{R}^{N_1} = \text{im}(\mathbf{B}_1^T) \oplus \text{ker}(\mathbf{L}_1) \oplus \text{im}(\mathbf{B}_2)$$

$$\mathbf{f}_1 = \mathbf{f}_G + \mathbf{f}_H + \mathbf{f}_C$$

$$\mathbb{R}^{N_0} \equiv \operatorname{im}(\mathbf{B}_1) \oplus \operatorname{ker}(\mathbf{L}_0)$$





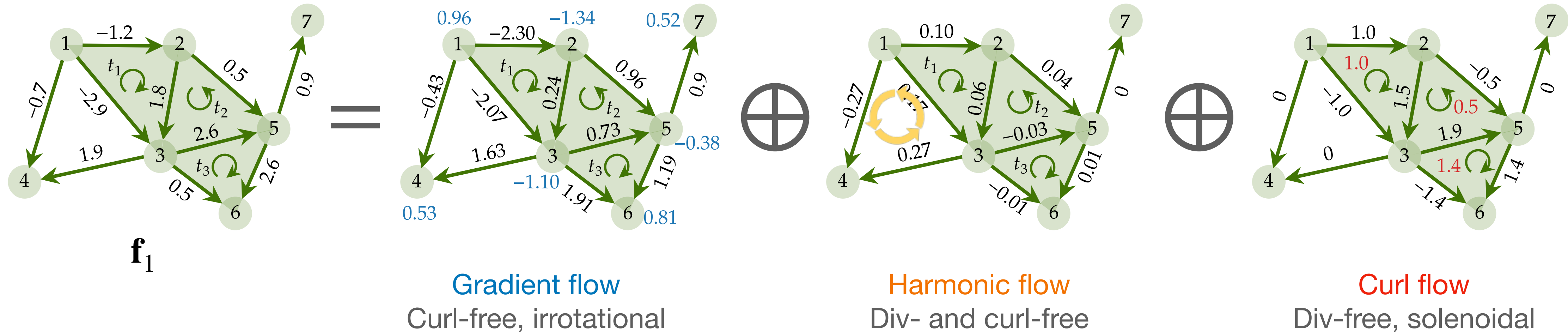


Hodge decomposition

Lovász et al. 2004; Lim et al. 2020

$$\mathbb{R}^{N_1} = \text{im}(\mathbf{B}_1^T) \oplus \text{ker}(\mathbf{L}_1) \oplus \text{im}(\mathbf{B}_2)$$

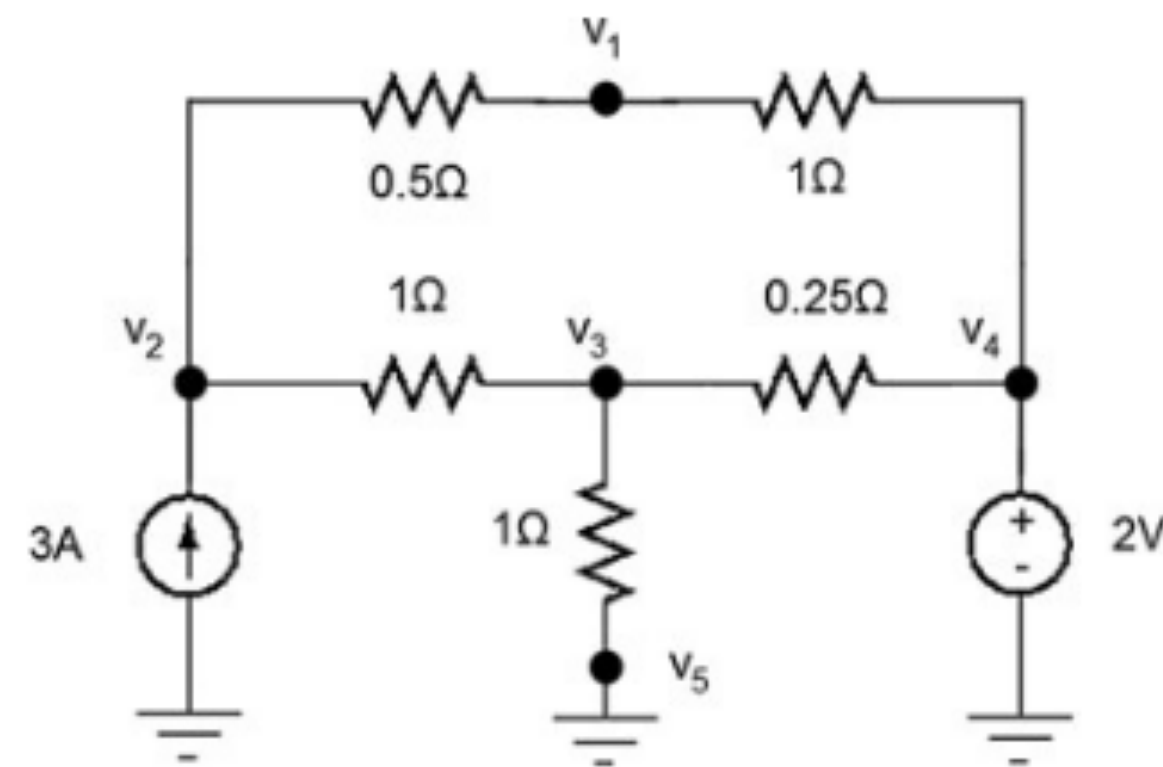
$$\mathbf{f}_1 = \mathbf{f}_G + \mathbf{f}_H + \mathbf{f}_C$$



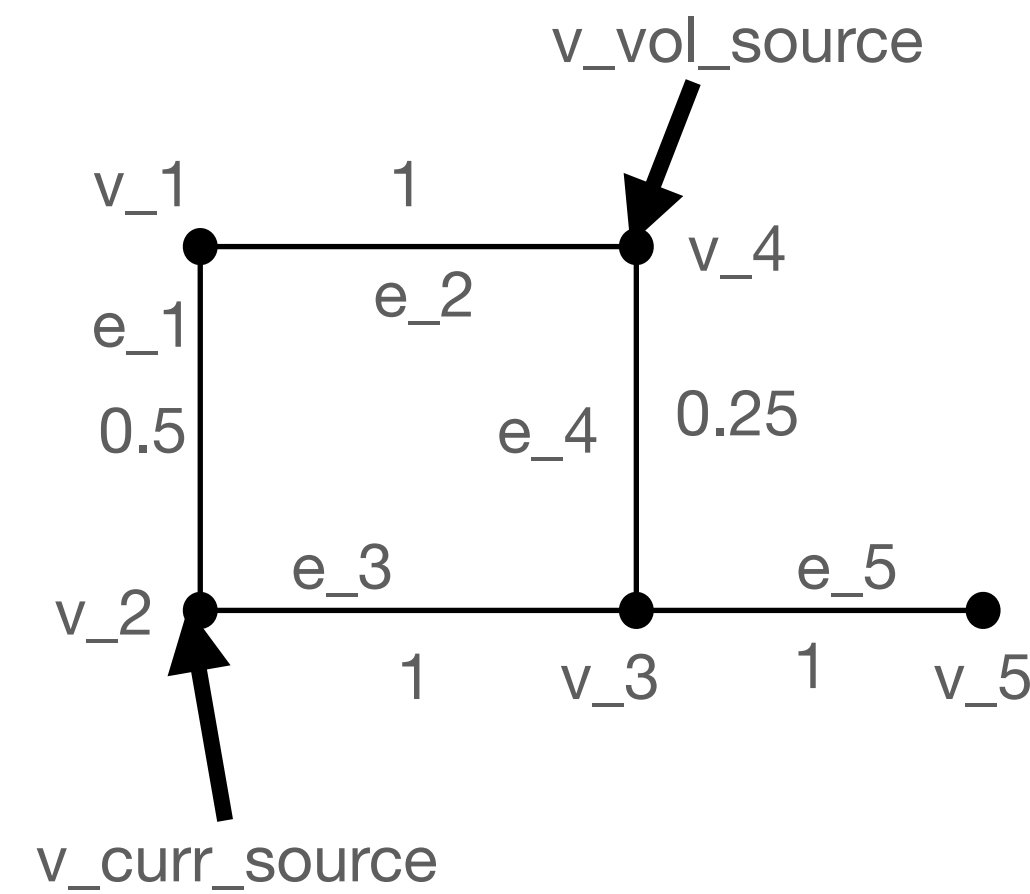
- This holds for the space of nodes, edges, triangles ... and **almost everywhere**
- What is the case for nodes? The harmonic space $\sim$ #connected components

$$\mathbb{R}^{N_0} = \text{im}(\mathbf{B}_1) \oplus \text{ker}(\mathbf{L}_0)$$

A Circuit toy example



(Grady et al. 2010)



Voltage (Kirchhoff's voltage law)
gradient flow, curl-free

$$\mathbf{B}_1 = \begin{pmatrix} 1 & 1 & 0 & 0 & 0 \\ -1 & 0 & 1 & 0 & 0 \\ 0 & 0 & -1 & 1 & 1 \\ 0 & -1 & 0 & -1 & 0 \\ 0 & 0 & 0 & 0 & -1 \end{pmatrix} \quad \mathbf{v}_{vol} = \begin{pmatrix} v_1 \\ v_2 \\ v_3 \\ 2 \\ 0 \end{pmatrix}$$

Current (Kirchhoff' current law)
curl flow, div-free